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Tsi12P

Tenkasi District

Common Quarterly Examination - September 2025

Standard 12

PHYSICS

Time Allowed: 3.00 Hours Maximum Marks: 70

PART - I

Choose the best answer: **15×1=15**

1) If 20J of work is done in bringing a negative charge towards a positive charge, then the electrostatic potential energy will be _____.
a) 20J b) −20J c) 10J d) −10J

2) A parallel plate capacitor stores a charge 'Q' at a voltage 'V'. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?
a) Capacitance b) Charge
c) Voltage d) Energy density

3) In India electricity is supplied for domestic use at 220V. It is supplied at 110V in USA. If the resistance of a 60W bulb for use in India is R, the resistance of a 60W bulb for use in USA will be _____.
a) R b) 2R c) $\frac{R}{4}$ d) $\frac{R}{2}$

4) A piece of copper and another of germanium are cooled from room temperature to 80K. The resistance of
a) each of them increases
b) each of them decreases
c) copper increases and germanium decreases
d) copper decreases and germanium increases

5) The value of Bohrmagneton which is used to measure atomic magnetic moments is _____.
a) 9.27×10^{24} Am^2 b) 9.27×10^{-24} Am^2
c) 8.78×10^{10} Am^2 d) 8.78×10^{-10} Am^2

6) The potential energy of magnetic dipole whose dipole moment is $P_m = (0.5\hat{i} + 0.4\hat{j})$ Am^2, kept in an uniform magnetic field $B = 0.2\hat{i}$T.
a) −0.1J b) −0.8J c) 0.1J d) 0.8J

7) Power delivered by Lorentz force on the particle of charge q moving with velocity V is _____.
a) Zero b) F . V c) $qV^2B\sin\theta$ d) infinity

8) A step-down transformer reduces the supply voltage from 220V to 11V and increase the current from 6A to 100A. Then its efficiency is _____.
a) 1.2 b) 0.83 c) 0.12 d) 0.9

9) In a series RL circuit, the resistance and inductive reactance are same. Then the phase difference between the voltage and current in the circuit is _____.
a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{6}$ d) Zero

10) The truth of Lenz's law can be established on the basis of the _____.
a) Law of conservation of momentum
b) Law of conservation of angular momentum
c) Law of conservation of energy
d) Law of conservation of isospin

11) Which of the following is an electromagnetic wave?
a) α rays b) β rays c) γ rays d) All of them

12) Fraunhofer lines are an example of _____ spectrum.
a) line emission b) line absorption
c) band emission d) band absorption

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13) If the amplitude of the magnetic field is 3×10^{-6} T, then amplitude of the electric field for a electromagnetic waves is _____.
a) 100 Vm^{-1} b) 300 Vm^{-1} c) 600 Vm^{-1} d) 900 Vm^{-1}

14) Stars twinkle due to
a) reflection b) total internal reflection
c) refraction d) polarization

15) For the light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is _____.
a) 30° b) 45° c) 60° d) 90°

PART - II

i) Answer any six questions. **6×2=12**
ii) Question No. 24 is compulsory.

16) The electric field lines never intersect. Justify.
17) State Joule's law of heating.
18) Define current density.
19) Compute the magnitude of the magnetic field of a long straight wire carrying a current of 1A at a distance of 1m from it.
20) What is meant by hysteresis?
21) Write any two differences between stepup transformer and step down transformer.
22) What are electromagnetic waves?
23) Why does sky appear blue?
24) A straight conducting wire is dropped horizontally from a certain height with its length along east-west direction. Will an emf be induced in it? Justify your answer.

PART - III

i) Answer any six questions. **6×3=18**
ii) Question No. 33 is compulsory.

C. Christo
MA.B.Ed
Tenkasi

25) Obtain Gauss law from Coulomb's law.
26) State and explain Kirchhoff's rules.
27) Discuss the conversion of galvanometer into voltmeter.
28) Compare dia, para, ferro magnetism.
29) How will you induce an emf by changing the area enclosed by the coil?
30) Mention the various energy losses in a transformer.
31) Write any six properties of electromagnetic waves.
32) Derive the relation between f and R for a spherical mirror.
33) Light travels from air into a glass slab of thickness 50 cm and refractive index 1.5.
a) What is the speed of light in the glass slab?
b) What is the time taken by the light to through the glass slab?
c) What is the optical path of the glass slab?

PART - IV

Answer all the questions: **5×5=25**

34) a) Derive the expression for resultant capacitance, when capacitors are connected in series and parallel.
(OR)
b) Derive an expression for electrostatic potential due to an electric dipole.

35) a) Describe the microscopic model of current and obtain microscopic form of ohm's law.
(OR)
b) Explain the determination of internal resistance of a cell using voltmeter.

36) a) Derive the expression for the force between two parallel current carrying conductors.
(OR)
b) Discuss principle, construction and working of cyclotron in detail.

37) a) Show that the mutual inductance between a pair of coils is same. $[M_{12} = M_{21}]$
(OR)
b) Explain the working of a single phase AC generator with necessary diagram.

38) a) Write down Maxwell equations in integral form, and explain them.
(OR)
b) Derive the mirror equation.

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